



# A systematic review of metaheuristic based feature selection strategies for cyber-attack detection in the IIoT

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## Abstract

The Industrial Internet of Things (IIoT) has been spreading across all fields of applicable environments, one of the most crucial ones being the industrial environment. IIoT is the integration of the internet within various industrial fields such as smart manufacturing, supply chain optimization, and predictive maintenance. These applications require two key features to be efficient, that is, real-time processing and connection security and reliability. From this standpoint arise the studies of cyberattack detection in industrial environments. Many methodologies have approached this field using traditional or hybridized machine learning or deep learning algorithms. In this review paper, we explore 36 of the most recent cyber-attack detection systems using metaheuristic models, mainly metaheuristic feature selection (MFS) algorithms. Additionally, we also explore hybrid models of metaheuristics and machine learning or deep learning models that are used to increase the accuracy of the models on various benchmark datasets. Our SLR separates the MFS utilized in this field into four main types, including Swarm Intelligence (SI), Evolutionary Algorithms (EA), Physics-Based (PHY), and Human-Behavior-Inspired (HBI). Our findings showed that SI-MFS dominates the field, with 25/36 case studies proposing it, while EA was proposed in 3/36 and PHY and HBI were each proposed in 2/36. We also demonstrate the most effective methodologies, such as FS-ID, MFS-D, and Novel Hybrid MFS. We also outline potential open challenges and gaps that require resolution.

**Keywords** Metaheuristic algorithms · Feature selection algorithms · Intrusion detection · IoT · IIoT · Machine learning · Deep learning

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## Abbreviations

|           |                                                                         |
|-----------|-------------------------------------------------------------------------|
| AAFSSO    | Asymmetrical Artificial Fish Swarm Optimization                         |
| ABC       | Artificial Bee Colony                                                   |
| AE        | Autoencoder                                                             |
| AFSA      | Artificial Fish Swarm Algorithm                                         |
| AI        | Artificial Intelligence                                                 |
| ANN       | Artificial Neural Network                                               |
| AQUO      | Aquila Optimizer                                                        |
| ASFOA     | Adaptive Sunflower Optimization Algorithm                               |
| BA        | Bat Algorithm                                                           |
| BBFS      | Bowerbird Courtship Feature Selection                                   |
| BGSA      | Binary Gravitational Search Algorithm                                   |
| BGWO      | Binary Grey Wolf Optimization                                           |
| BSA       | Bird Swarm Algorithm                                                    |
| CASA      | Channel and Spatial Attention                                           |
| CAT-S     | Cellular Automata Tabu Search                                           |
| CNN       | Convolutional Neural Network                                            |
| CRNN      | Cascaded Recurrent Neural Network                                       |
| CSA       | Crow Search Algorithm                                                   |
| CVS       | Chaotic Vortex Search                                                   |
| DFFA      | Diversity-Enhanced Firefly Algorithm                                    |
| DHOA      | Deer Hunting Optimization Algorithm                                     |
| DL        | Deep Learning                                                           |
| DNN       | Deep Neural Network                                                     |
| DRNN      | Deep Recurrent Neural Network                                           |
| DTO       | Dipper-Throated Optimizer                                               |
| EA        | Evolutionary Algorithms                                                 |
| EFPADD    | Elderly Fall Prediction and Detection                                   |
| EHO       | Elephant Herding Algorithm                                              |
| ETM       | Embedded Transformer Monitoring                                         |
| FFA       | Firefly Algorithm                                                       |
| FLN       | Fast Learning Network                                                   |
| FR-CNN    | Fast Recurrent Convolutional Neural Network                             |
| FS        | Feature Selection                                                       |
| FS-ID     | Feature Selection—Intrusion Detection                                   |
| FWA       | Fireworks Algorithm                                                     |
| GA        | Genetic Algorithm                                                       |
| GNB       | Gaussian Naive Bayes                                                    |
| GOA       | Grasshopper Optimization Algorithm                                      |
| GSA       | Gravitational Search Algorithm                                          |
| GTO       | Gorilla Troops Algorithm                                                |
| GWO       | Gray Wolf Optimizer                                                     |
| HADE-DRNN | Heuristic Adaptive Differential Evolution-Deep Recurrent Neural Network |
| HAEMPSO   | Hybrid Autoencoder and Modified Particle Swarm Optimization             |
| HBI       | Human Behavior Inspired                                                 |

|         |                                               |
|---------|-----------------------------------------------|
| HGS     | Hunger Games Search                           |
| HHO     | Harris Hawk Optimizer                         |
| IBPSO   | Improved Binary Particle Swarm Optimization   |
| ID      | Intrusion Detection                           |
| IDS     | Intrusion Detection System                    |
| IIoT    | Industrial Internet of Things                 |
| IoT     | Internet of Things                            |
| KNN     | K-Nearest Neighbors                           |
| LR      | Logistic Regression                           |
| MFS     | Metaheuristic Feature Selection               |
| MFS-DL  | Metaheuristic Feature Selection-Deep Learning |
| MIA     | Mathematically Inspired Algorithm             |
| ML      | Machine Learning                              |
| PHY     | Physics Based                                 |
| PSO     | Particle Swarm Optimization                   |
| RF      | Random Forest                                 |
| SA      | Self Attention                                |
| SFO     | Sailfish Optimization                         |
| SFOA    | Sunflower Optimization Algorithm              |
| SI      | Swarm Intelligence                            |
| SMOTE   | Synthetic Minority Oversampling Technique     |
| SOA     | Seagull Optimization Algorithm                |
| SSA     | Salp Swarm Algorithm                          |
| SSO     | Sparrow Search Optimization                   |
| SVM     | Support Vector Machine                        |
| SWO     | Spider Wasp Optimization                      |
| TS      | Tabu Search                                   |
| TSO     | Tuna Swarm Optimization                       |
| VP      | Voting Perceptron                             |
| WOA     | Whale Optimization Algorithm                  |
| XGBoost | Extreme Gradient Boosting                     |

## 1 Introduction

Industrial Internet of Things (IIoT) security has become an essential requirement in any IIoT environment ever since cyberattacks began evolving through attackers, and its integration with industrial applications further cemented this necessity in industrial fields (Mekala et al. 2023). One of the most effectively known ways that have been used to mitigate these cyberattacks is metaheuristic algorithms (Hussain et al. 2019), specifically metaheuristic feature selection (MFS) algorithms (Mohammed et al. 2024; Li et al. 2025). Metaheuristics are mathematical optimization problems that simulate nature or human intelligence. Metaheuristics can be sorted into two basic categories: single-solution-based (SSB) (Jaddi and Abdullah 2020) and population-based (PB) (Samsuddin et al. 2018; Velasco et al. 2024). SSB's initiate with a single unique solution that gets modified while iterating. PB, on the other hand, creates many solutions and then optimizes with the number of iterations. PB

algorithms are advantageous over SSB for dodging falling for local optima and multi-solution-assisted in-depth examination of search space (Agrawal et al. 2021). Furthermore, metaheuristics can also be subdivided based on their behavior into four categories:

- **Evolution-Based:** derived from the natural evolution. The most famous one is the Genetic Algorithm (GA). They evolve through selection, crossover, and mutation (Maier et al. 2019).
- **Swarm intelligence (SI) based:** derived from animals' social behavior. Famous SI algorithms include particle swarm optimization (PSO), firefly algorithm (FFA), ant colony optimization algorithm (ANO), grey wolf optimizer (GWO), bat algorithm (BA), and whale optimization algorithm (WOA) (Brezočník et al. 2018; Tang et al. 2021; Zedadra et al. 2018; Khare et al. 2023).
- **Physics-Based:** derived out of our universe's physics rules, like simulated annealing, gravitational search algorithm (GSA), and harmony search (Seyyedabbasi 2024).
- **Human Behaviour Related:** Purely derived from human behavior, like tabu search, teaching learning-based (TLB), and league championship (Rai et al. 2022).

Feature selection metaheuristics (MFS) algorithms are search algorithms for optimization techniques used to propose new feature subsets to be used for evaluating algorithms in a dataset by measuring the effectiveness of different feature subsets (Dissanayake 2021). For cyberattacks, metaheuristic feature selection (FS) algorithms followed by a machine or deep learning model have been a common approach to detect them, more often in IoT (Hu et al. 2024). FS algorithms consist of three different types: Filter Methods (Arya and Gupta 2023; Kumar et al. 2021; Lyu et al. 2023; Soe et al. 2020), Wrapper Methods (Maldonado et al. 2022; Maseno and Wang 2024), and Embedded Methods (Kasongo and Sun 2020; Agrawal et al. 2021). Most of the FS algorithms used in this field are said to be wrapper-based FS, consisting of three components: search strategy, predictor function, and evaluation or fitness function.

IIoT datasets, specifically IIoT intrusion detection (ID) datasets, as IIoT devices generate massive quantities of data from numerous sensors and devices. Therefore, IIoT intrusion detection datasets also exhibit high dimensionality and noise. Therefore, feature selection (FS) can be applied to enhance the quality of the dataset for testing intrusion detection (ID) algorithms, improving their accuracy and reducing their time complexity. Furthermore, this paper will discuss multi-feature selection (MFS), an excellent optimization methodology for this purpose.

This paper examines 36 modern case studies that use individual or hybrid metaheuristic algorithms for feature selection of intrusion detection data in IIoT and their effectiveness in improving machine and deep learning classification in IIoT domains. The contributions of this work include the following.

- The work effectively illustrates the most efficient metaheuristic methodologies for intrusion detection in IIoT.
- Provides a comprehensive technical analysis of improved feature selection and improved intrusion detection and points out the research gaps that are yet to be fulfilled by metaheuristics and key trade-offs.
- Identify open challenges, dominant trends, and industrial practical feasibility and bar-

riers to adoption.

Furthermore, we inquire about the methodological questions below:

1. MQ1: What types of metaheuristic algorithms have been employed for feature selection in intrusion detection systems within IIoT environments?
2. MQ2: Which metaheuristic-based feature selection algorithms demonstrate the highest performance in IIoT intrusion detection systems?
3. MQ3: What are the most commonly used benchmark datasets for evaluating intrusion detection in IIoT?
4. MQ4: What novel metaheuristic models and novel hybrid metaheuristic models have been proposed for intrusion detection in IIoT?
5. MQ5: What types of intrusion detection algorithms are integrated with metaheuristic-based feature selection methods in IIoT?
6. MQ6: What are the most frequently reported limitations and research gaps in existing MFS-based IIoT intrusion detection approaches?
7. MQ7: What future research directions and open challenges have been identified in employing metaheuristic-based feature selection for IIoT intrusion detection?

The remainder of this paper is divided as follows. Section 2 highlights similar works to ours. 3 a summary of the selected research methodologies. Section 4 is a visualization of the taxonomy of MFS-based IDS algorithms. Section 5 presents technical analysis and discussion. Section 6 presents open challenges. And finally, Sect. 8 concludes the paper.

## 2 Related work

This section highlights related works, including recent systematic literature reviews similar to our SLR and case studies that were ultimately excluded from this study.

### 2.1 Similar review papers

This subsection highlights key reviews that are similar to our work and compares them with it. This comparison can be seen in Table 1

Hu et al. (2024) presented a comprehensive analysis of nature-inspired and meta-heuristic algorithms for designing intrusion detection systems in cloud/edge and IoT environments. This paper provides a comprehensive review of nature-inspired algorithms applied to intrusion detection systems (IDS) in cloud, edge, and IoT environments. It focuses on the challenges of designing IDS for these ecosystems and evaluates various metaheuristics, such as PSO, GWO, and ALO, for feature selection and optimization. The datasets used include NSL-KDD, CICIDS-2017, and UNSW-NB15, which are general-purpose datasets for network intrusion detection. While the paper discusses diverse FS and ID algorithms that achieved high accuracy and detection rates, it primarily addresses general IoT and cloud/edge environments, with a limited focus on the unique challenges of IIoT.

A second review paper, which included MFS algorithms, is Mallidi and Ramisetty (2025) performed an SLR of ML and DL techniques for evolving cyber threat detection in IoT

**Table 1** Comparison of related works with this study

| Study                         | Publication year | Primary focus                                                                           | FS role                                   | IIoT-specific                     | Key distinction                                                                                                                                          |
|-------------------------------|------------------|-----------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hu et al. (2024)              | 2024             | Nature-inspired and meta-heuristic algorithms for IDS in cloud/edge and IIoT            | Metaheuristic FS for optimization         | Limited                           | General IIoT and cloud/edge environments                                                                                                                 |
| Mal-lidi and Ramisetty (2025) | 2024             | ML/DL approaches for IIoT intrusion detection with feature selection and data balancing | Metaheuristic FS among broader techniques | Limited                           | Systematic review of ML/DL with feature selection as one aspect                                                                                          |
| Sadeghian et al. (2025)       | 2025             | Nature-inspired algorithms for feature selection (2018–2022)                            | Primary focus on metaheuristic FS         | No                                | General cybersecurity/IDS applications without IIoT focus                                                                                                |
| Souri et al. (2024)           | 2024             | Cloud-based cyber-threat detection for hyper-automation in IIoT                         | ANOVA-based FS (statistical)              | Yes—Industry 5.0 hyper-automation | Cloud-based ensemble approach with statistical FS, limitations in detecting exploits                                                                     |
| Our Study                     | Nov/2025         | <b>Cyber-attack detection in IIoT using metaheuristic feature selection (2018–2025)</b> | <b>Primary focus on MFS algorithms</b>    | <b>Yes—Dedicated</b>              | <b>Explicit IIoT focus addressing industrial requirements (real-time processing, security), MFS as main methodology, industrial cyber-attack context</b> |

ecosystems, focusing majorly on FS and data balancing optimization mechanisms. They examined a diversity of metaheuristic FS (MFS) models, such as GA, PSO, ACO, and relatively novel hybridized methods. They also examined how datasets like NSL-KDD, CICIDS-2017, UNSW-NB15, ToN-IIoT, and BoT-IIoT were being functionalized to train models in this context. This SLR, however, while it includes MFS algorithms, is not designated specifically for them nor specifically designated for their usage in an IIoT context.

Additionally, Sadeghian et al. (2025) The 2025 research investigated how nature-inspired algorithms revolutionized feature selection (FS) from 2018 to 2022, examining techniques ranging from GAs to PSO and ACO. These computational approaches, which simulate biological and physical processes, excel at transmitting complex feature spaces and have proven particularly effective in cybersecurity applications like IDS. The study reveals that hybrid methods combining different selection strategies—such as FFA, GSA, and ACO—are consistently defeating regular techniques on established datasets like KDD-CUP 99 by discovering optimal feature subsets that upgrade detection rates while minimizing computational overhead. While these algorithms successfully address challenges like local optima and feature redundancy, the researchers argue that algorithm selection should be designated for specific dataset traits and application necessities rather than any total assumption of a single approach that is projected to excel universally across all domains.

Finally, Souri et al. (2024) proposed a cloud-based cyber-threat detection architecture for hyper-automation procedures in IIoT environments. The research introduces a hybrid Ensemble Bagged Trees Detection (EBTD) model with an ANOVA (Analysis of Variance)-

based FS method to identify malicious patterns in industrial systems. The EBTD algorithm outperforms other ML approaches, including SVM, decision tree, and ANN. It scores a near-perfect detection accuracy of 99.51% on UNSW-NB15 and 99.86% on NSL-KDD datasets, while majorly reducing computation time. The architecture is specifically designated for hyper-automation applications in Industry 5.0 environments, focusing on cloud-based security for industrial communication systems. One limitation of the approach is that, although it performs well on benchmark datasets, the precision and recall results for detecting exploits are not sufficiently high, indicating limitations in the FS strategy for certain types of attacks.

## 2.2 Excluded case studies

This subsection highlights similar case studies that were eventually excluded from this work.

First, Sangaiah et al. (2023) proposed a Hybrid Ant-Bee Colony Optimization (HABCO) for FS with ID classifiers in a cloud of things (CoT). They addressed the pressing security issues of CoT, which require robust monitoring tools for identifying malicious behaviors involving a large number of connected devices. By formulating FS as an optimization problem, HABCO integrates ACO and ABC algorithms sequentially to select the optimal feature attributes that discriminate between benign and malignant network behavior. The proposed approach was evaluated on the KDD Cup 99 data set using seven different classifiers, and it achieved higher accuracy rates in the range of 2.87–4.63% compared to other state-of-the-art approaches. The work demonstrates a metaheuristic approach to optimizing security in CoT systems, where traditional IDS often bottleneck in these environments due to the vast number and complex nature of today's IoT-based devices and the resources in the cloud. While this work uses an MFS model and covers most of our inclusion criteria, it focuses on CoT rather than IIoT; we excluded it from our work. Another work that has a similar inclusion–exclusion situation is Davahli et al. (2020a) and Davahli et al. (2020b) proposed a novel hybridization of an evolutionary and an SI metaheuristic, first for FS in WSNs in IoT and second in IoT wireless networks. The hybrid MFS model followed sequentially by an SVM classifier was experimented on with the real-world AWID dataset and scored an accuracy of 99.09%. The paper details time complexity to be lightweight, with an FS time of 1147.156 s and a detection time of 10.906 s. However, this paper lacks details on overfitting analysis and is limited with its experiments to a single dataset. Despite the impressive novel MFS, the paper focuses on wireless networks in relation to the IoT approach rather than IIoT; therefore, it was excluded as a case study.

Many resources proposed MFS for ID in IoT but were too general to be included in IIoT-specific terms. For instance, Prasath and Perumal (2019) proposed a hybrid FS-ID algorithm that mixes PSO with a Bayes net classifier. The model, nicknamed MHBNC, was tested on the NSL-KDD dataset, scoring an overall accuracy of 82.99% over 4 attack types and normal data. The paper, like many of its peers, does not detail time complexity nor overfitting analysis. Furthermore, the paper presents this model for general use on network intrusion detection (ID) or Internet of Things (IoT) applications but lacks specific details for Industrial Internet of Things (IIoT) contexts. Additionally, its sole reliance on the NSL-KDD dataset raises concerns about its applicability as a case study in more specialized environments. For these reasons, it was excluded as a case study and included only as a resource. Another example of these is Alamiedy et al. (2020) proposed a NIDS utilizing

an FS-ID sequence of a multi-objective GWO and an SVM. The testing was conducted on the NSL-KDD dataset and had 4 attack types to classify, those being DoS, Probe, R2L, and U2R. The model scored accuracies of 93.64%, 91.02%, 57.73%, and 53.70% on the aforementioned attacks, respectively. This research, however, did not address either time complexity or overfitting. Furthermore, it targeted general IoT usage and was only tested on a general dataset, that is, NSL-KDD. Thus, it was excluded as a case study and included only as a related work. An additional example is that Karthikeyan et al. (2025) integrated an MFS with ensemble learning techniques for cyber threat recognition in IoT. For this, an adaptive Harris hawk optimizer (HHO) was applied on the CICIDS-2017 and NSL-KDD datasets along with a hybrid bidirectional gated recurrent unit (GRU) and WAE (Wasserstein autoencoder) and DBN (deep belief network), with the FS-ID executed sequentially. The proposed model achieved accuracies of 99.44% and 98.85% on the aforementioned datasets, respectively. A notable weakness of this paper is that it lacks a detailed description of time complexity and only states that the model is computationally efficient. Furthermore, this paper lacks relevance to IIoT, and thus was not considered to be a case study.

Other case studies targeted IIoT but applied a non-metaheuristic FS while utilizing meta-heuristics for hyperparameter optimization rather than FS. For instance,

Khafaga et al. (2023) presented a voting classifier and metaheuristic optimization for network intrusion detection. This paper proposes a NIDS with a voting perceptron (VP) based at its core on three classifiers: NN, RF, and KNN. The VP's parameters were enhanced using a modified whale optimization technique (WOA) led by a dipper-throated optimizer (DTO), where both WOA and DTO are metaheuristics. The model produced was executed on the RPL-NIDS17 dataset. SMOTE and LSH techniques were utilized to balance the dataset's traffic. FS was applied using an unspecified algorithm. The NN, RF, and KNN achieved a total accuracy of 95.1%, with an AUC of 99% and an MSE of 2.50E-08. These results outperformed the previous approaches; however, this research lacks a detailed explanation of the feature selection algorithm used. It also could have used a secondary dataset like CICIDS-2017. Furthermore, the papers include a detailed analysis of its models' time complexity.

Another example of these is Asif (2025), who presented robust ID in IoT using a stacked ensemble network with GA and DNN. This research aims at detecting various cyber threats in different network environments. It created a novel FS algorithm for the channel and spatial attention (CASA) technique, which enables focus on distinguished sets of features while avoiding useless or noisy data. CASA was followed by a hybrid CNN that is optimized with a GA meta-learner. Four datasets were used: Edge-IIoTset, IoT\_Malware, the Blended Malware dataset, and UNSW-NB15. It got accuracies of 99.71%, 96.17%, 96.37%, and 99.15%, respectively. The weakness of this paper is that it signified a computational complexity trade-off, as the ensemble technique raises overhead with no valuable performance benefits, in addition to a lack of real-world testing. Furthermore, another example lies in Shukla et al. (2025), who suggested a feature selection method and an MHO followed by a hybridized DL model for IoT FS-HPO-ID. It utilized a Levy flight mathematical technique based on the grasshopper optimization algorithm (GOA) for hyperparameter optimization (HPO) and a kernel principal component analysis (K-PCA) for FS. It was followed sequentially by a hybrid DNN-LSTM and tested on the NSL-KDD, TON-IoT, and CIC-IDS 2017, scoring accuracies of 98.67%, 97.22%, and 98.54%, respectively. Training time with the proposed



method was 10,392 sec (NSL-KDD), 10,031 sec (TON-IoT), and 11,007 sec (CIC-IDS 2017).

Which is reasonable given its DL, and it is faster than base forms of CNN, DNN, LSTM, CNN-LSTM, or DNN-LSTM. The research mentioned functionalizing adaptive synthetic oversampling and 5-fold cross-validation to remove any overfitting. The paper has relevance to IIoT through the TON-IoT dataset and also industrial mentions in its text. However, the FS method used, K-PCA, is not metaheuristic. And metaheuristics in this research were only used for hyperparameters.

Another similar situation like this also includes (Chen and Mir 2025), who suggested an FS method and an MHO followed by a hybridized DL model for IIoT FS-HPO-ID. It utilized a Reptile Search Algorithm combined with an Improved Whale Optimization Algorithm (RSA-IWOA) for hyperparameter optimization and Principal Component Analysis (PCA) for FS. It was followed sequentially by a hybrid CNN-LSTM and tested on the UNSW-NB15, TON-IoT, and NCTUKM-IIOT datasets, scoring accuracies of 97.6%, 98.9%, and 98.7%, respectively. Training time with the proposed method was 50 min (with PCA and 45 features) compared to 78 min (without PCA and all features), producing efficiency enhancements through dimensionality reduction. The research focused specifically on IIoT environments and utilized IIoT-specific datasets, including NCTUKM-IIOT. The paper has strong relevance to IIoT through its focus on industrial control systems, use of the TON-IoT and NCTUKM-IIOT datasets, and extensive discussion of industrial applications. However, the FS method used, PCA, is not metaheuristic. And metaheuristics in this research were only used for hyperparameters.

Other papers excluded for different reasons include (Bhattacharyya et al. 2020), who innovate a novel MFS method by hybridizing the Mayfly algorithm and Harmony Search (MA-HS). This is a novel cross-category hybridization since MA is an SI algorithm while HS is physics-based. This model, followed by classification, achieved high accuracies across many datasets. The research additionally provides a comparative analysis of the proposed model against other models such as PSO, GSA, SSA, and HHO. However, this research presents it as a general-purpose FS model and not for IDS or IoT usage. For this reason, it was excluded as a case study. Benmessahel et al. (2019) propose utilizing SI locust swarm optimizer (LSO) and evolutionary neural networks for weight training and ID in IoT. The model scored detection accuracies of 95.42% and 94.02% on the UNSW-NB15 and NSL-KDD, respectively. Benmessahel et al. (2018) also proposed a similar idea but utilizing the physics-based multiverse optimizer (MVO) and evolutionary neural networks for weight training instead of LSO. This variation scored detection accuracies of 96.61% and 98.21% on the UNSW-NB15 and NSL-KDD, respectively. Both of these papers are a great example of the evolution of metaheuristic usages to be used beyond just FS. Both papers, however, do not explicitly perform FS; therefore, they were excluded as case studies.

After a careful literature review, we deduced that a research gap exists: there is no comprehensive summary of modern metaheuristic algorithms that are functionalized for feature selection of IIoT intrusion-detection data. Thus, we held this systematic literature review to identify, compare, and synthesize these methods.

### 3 Selected research methodology

This section outlines the key methodologies used in attack detection using IIoT metaheuristics, focusing on feature selection (FS) methods and classification models. We examine the most utilized and practical approaches in IDS for IIoT (IIoT).

#### 3.1 Summary of methodologies

This subsection outlines the most utilized MFS-Based methodologies for IIoT attack detection.

##### 3.1.1 Feature selection techniques and classification models

In 36 analyzed case studies, a total of 26 different types of metaheuristic algorithms were proposed for feature selection. 23 of these 26 were population-based algorithms, while 3 were single-solution-based.

Out of 36 case studies, 27 case studies applied single MFSs individually, while 9 hybridized two or more MFSs. 30 case studies applied FS-ID sequentially, while 6 case studies hybridized the MFS algorithm with the ID into a single algorithm. 32 case studies utilized population-based metaheuristics, while only 4 used single-solution-based ones (two TS, CVS, and GRASP). 24 case studies utilized swarm intelligence (SI) metaheuristics, while 4 utilized evolutionary (all genetic algorithms), 2 used physics-based (FWA and CVS), 2 used human behavior-based (TS), and 3 case studies proposed hybrid MFS between the origin categories. Two of the single-solution cases utilized the human-based behavior (HBI) tabu search (TS) algorithm, while the other two used the physics-based chaotic vortex search (CSV) and the unique GRASP metaheuristic, which is mathematically inspired. Table 2 summarized all algorithms used and the number of times they were used individually or were hybridized with another MFS.

The most used evolutionary MFS algorithm was GA, with 5, three times individually and twice in novel hybrids, placing it as the most used MFS algorithm overall. Second overall are PSO and GWO with 4, with each being used 3 times individually and once in a novel hybrid, exactly matching. PSO and GWO are the most used SI algorithms overall, and within SI hybridization they only tie with FFA for 3, and then come WOA, AFSA, SSA, and GTO with 2 usages per MFS.

On the physics-based algorithms side, only two algorithms were used, one each, those being CVS and FWA. In HBI, only TS was used twice. All of the remaining algorithms were only used once overall, displaying a wide vision across the current popularity of these models. All of the FS algorithms used were wrapper-based methods, as metaheuristic optimizers are designated to search into the feature space to find optimized subsets based on classifier performance (Chandrashekar and Sahin 2014). For classification, RF was the most used ML classifier with 4/36 case studies, while deep learning's ANN, CNN, and RNN were tied with 6 usages each. Tables 3 and 4 summarize the number of usages per algorithm per category and the number of times it was used individually or was hybridized.

The trend heads towards sequentializing the FS-ID process rather than hybridizing it, with 30/36 of the case studies following this sequential order, and only 4/36 papers hybridizing the MFS with the IDS model, with the remaining two not specifying. Meanwhile, 10/36

**Table 2** Metaheuristic feature selection algorithm usage analysis

| Algorithm                                   | Total | Individual MFS | Hybrid with another MFS |
|---------------------------------------------|-------|----------------|-------------------------|
| Particle Swarm Optimization (PSO)           | 4     | 3              | 1                       |
| Genetic Algorithm (GA)                      | 5     | 3              | 2                       |
| Firefly Algorithm (FFA)                     | 3     | 3              | 0                       |
| Gray Wolf Optimizer (GWO)                   | 4     | 3              | 1                       |
| Tabu Search (TS)                            | 2     | 2              | 0                       |
| Whale Optimization Algorithm (WOA)          | 2     | 1              | 1                       |
| Sailfish Optimizer (SFO)                    | 1     | 0              | 1                       |
| Artificial Fisher Swarm Algorithm (AFSA)    | 2     | 1              | 1                       |
| Grasshopper Optimization Algorithm (GOA)    | 1     | 0              | 1                       |
| Bat Algorithm (BA)                          | 1     | 1              | 0                       |
| Tuna Swarm Optimization (TSO)               | 1     | 1              | 0                       |
| Bird Swarm Optimization (BSO)               | 1     | 0              | 1                       |
| Gorilla Troops Optimizer (GTO)              | 2     | 0              | 2                       |
| Crow Search Algorithm (CSA)                 | 1     | 1              | 0                       |
| Salp Swarm Algorithm (SSA)                  | 2     | 1              | 1                       |
| Seagull Optimization Algorithm (SOA)        | 1     | 0              | 1                       |
| Elephant Herding Optimization (EHO)         | 1     | 0              | 1                       |
| Harris Hawk Optimizer (HHO)                 | 1     | 1              | 0                       |
| Aquila Optimizer (AQUO)                     | 1     | 1              | 0                       |
| Deer Hunting Optimization Algorithm (DHOA)  | 1     | 1              | 0                       |
| Sunflower Optimizing Algorithm (SFOA)       | 1     | 1              | 0                       |
| Fireworks Algorithm (FWA)                   | 1     | 1              | 0                       |
| Chaotic Vortex Search (CVS)                 | 1     | 1              | 0                       |
| Gravitational Search Algorithm (GSA)        | 1     | 0              | 1                       |
| Spider Wasp Optimization (SWO)              | 1     | 0              | 1                       |
| Artificial Bee Colony (ABC)                 | 1     | 0              | 1                       |
| Bowebird courtship feature selection (BBFS) | 1     | 1              | 0                       |

**Table 3** Machine learning algorithm usage with MFS

| Algorithm                                            | Total | Indep. | Hybrid with another ID |
|------------------------------------------------------|-------|--------|------------------------|
| Random forest                                        | 4     | 4      | 0                      |
| XGBoost                                              | 3     | 3      | 0                      |
| K-nearest-neighbour                                  | 2     | 2      | 0                      |
| Split-point and attribute reduced classifier(SPAARC) | 1     | 1      | 0                      |
| K-means                                              | 1     | 0      | 1                      |
| Expert classifier approach (ECA)                     | 1     | 1      | 0                      |
| Bagging                                              | 1     | 0      | 1                      |
| Logistic regression (LR)                             | 1     | 0      | 1                      |
| LightGBM                                             | 1     | 0      | 1                      |

cases used hybrid MFS approaches, 25/36 used MFS individually and a single case, MFS was hybridized with a non-metaheuristic for FS (TS + Cellular Automata). 8/36 cases have used hybrid ID methods. 17/36 cases generally use hybrid algorithms of any type, meaning the likelihood of a case study using a hybrid algorithm is 47.2%.

**Table 4** Deep learning algorithms usage with MFS

| Algorithm                                     | Total | Indep. | Hybrid |
|-----------------------------------------------|-------|--------|--------|
| Convolutional Neural Network (CNN)            | 6     | 2      | 4      |
| Deep Neural Network (DNN)                     | 2     | 2      | 0      |
| Bidirectional Long Short-Term Memory (BiLSTM) | 2     | 0      | 2      |
| Long Short-Term Memory (LSTM)                 | 1     | 0      | 1      |
| Artificial Neural Network (ANN)               | 6     | 4      | 2      |
| Recurrent Neural Network (RNN)                | 6     | 4      | 2      |
| Autoencoder                                   | 2     | 1      | 1      |
| Capsule Network (CapsNets)                    | 1     | 0      | 1      |
| Deep Belief Network (DBN)                     | 1     | 0      | 1      |

**Table 5** Comparison of single versus hybrid feature selection algorithms

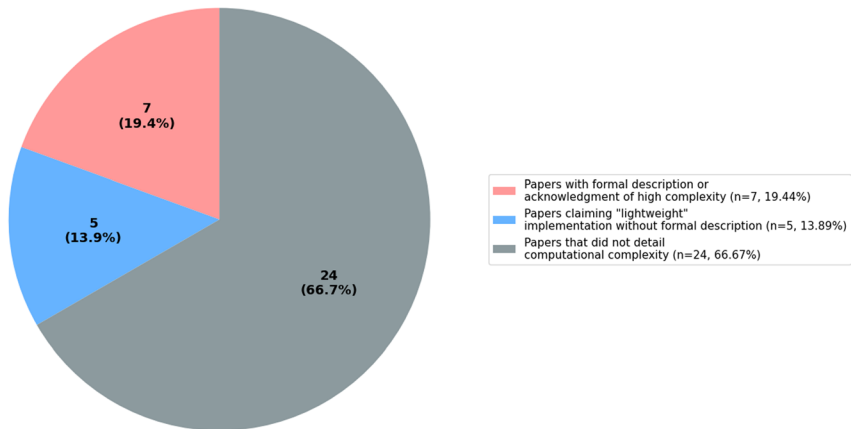
| Metric               | Single MFS algorithms                    | Hybrid MFS algorithms                               |
|----------------------|------------------------------------------|-----------------------------------------------------|
| Top performers       | HHO- (100%), FFA (99.92%), IBSWO (99.9%) | DOFFA (100%), Co-ati-GWO (99.91%), HAEM-PSO (99.9%) |
| Mean accuracy        | 97.27%                                   | 97.94%                                              |
| Median accuracy      | 99.5%                                    | 98.96%                                              |
| Minimum accuracy     | 79.11% (PSO)                             | 91.6% (SAEHO)                                       |
| Maximum accuracy     | 100% (HHO)                               | 100% (DOFA)                                         |
| Number of algorithms | 26                                       | 10                                                  |

**Table 6** Comparison of single versus hybrid intrusion detection algorithms

| Metric                 | Single ID algorithms                                   | Hybrid ID algorithms                                                                          |
|------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Top performers         | Random forest (100%), RNN (100.00%), SPARRC (99.9962%) | IBSWO (GA-SWO) Hybrid (99.9%), PSO-KMeans (99.8%), CNN-BiLSTM with attention (ACBLST) (99.8%) |
| Mean accuracy          | 96.59%                                                 | 98.51%                                                                                        |
| Median accuracy        | 99.20%                                                 | 99.37%                                                                                        |
| Minimum accuracy       | 76.00% (CNN-DBN)                                       | 89.54% (NN-LSTM)                                                                              |
| Maximum accuracy       | 100.00% (random forest)                                | 99.90% (IBSWO (GA-SWO) hybrid)                                                                |
| Number of case studies | 28                                                     | 8                                                                                             |

This distribution reflects strategic methodological evolution in the field. Simple metaheuristics often encounter early convergence or optimization inefficiencies when applied to complex threat detection scenarios. The 47.2% of studies employing hybrid approaches demonstrate the value of combining complementary algorithmic strengths, with these hybrids scoring impressive accuracy improvements over their individual classical implementations. Table 5 quantifies the performance differential between hybrid and single metaheuristic approaches when paired with an intrusion detection classifier, while Table 6 quantifies the same comparison for the utilized ID algorithms.

However, this improvement in accuracy trades off with computational costs that can restrict the ease of deployment in resource-limited IIoT devices. Only 7/36 of the case studies actually mention their models' time complexity in detail, while 5/36 claimed lightweight and 24/36 did not detail it. Figure 1 clarifies this paradox.

**Distribution of Computational Complexity Analysis in Papers**

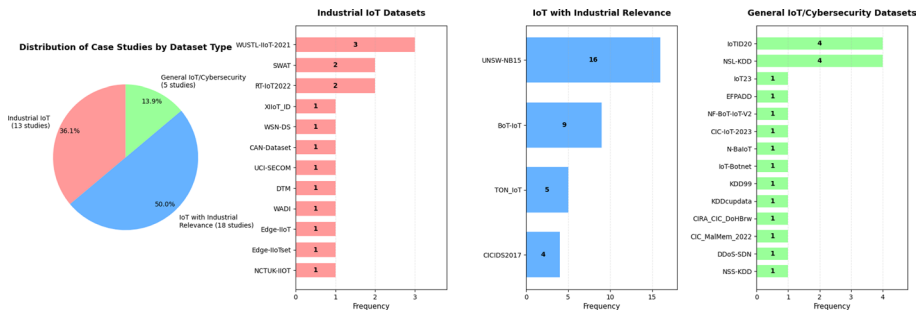
**Fig. 1** Percentage of time complexity mention in the case studies

This is a noteworthy observation of the disconnection between algorithmic sophistication and practical implementation. The low 7/36 or 19.4% indicates a crucial research gap, as understanding resource requirements is necessary to interpret theoretical performance into operating value. The lack of edge device validation is very problematic, as algorithms scoring well on research clusters may be absolutely impractical for deployment on actual IIoT devices.

Furthermore, there is the dataset mismatch issue. 13.9% of the research used IoT test-beds, while 50% of them used IoT datasets with some industrial relevance, and only 36.1% used IIoT-specific datasets. While the industrial relevance is strong, it adds up with the general IoT datasets to 63.9% non-industrial-specific datasets, which suggests many proposed solutions may not adequately address the unique issues of industrial environments, although they are still good proposals to try. The domination of TCP/IP-focused threat detection overlooks industrial protocol-specific threats (Modbus/TCP, EtherNet/IP), which have different statistical properties and necessitate designated feature engineering. Figure 2 sorts all datasets per category and lists the number of times each dataset was used.

Those datasets categories include attacks as follows:

- Industrial IoT (IIoT) Datasets
  1. Physical Process Manipulation
  2. SCADA/ICS Attacks
  3. Industrial Protocol Exploits (Modbus, DNP3, CAN)
  4. PLC/Controller Compromise
  5. Data Injection/False Readings
  6. DoS on Industrial Systems
- IoT Datasets with industrial relevance



**Fig. 2** Datasets categorized based on IoT general, IoT with IIoT relevance and IIoT specific and their usage amount

1. DDoS/Botnet Attacks
2. Network Intrusion
3. Port Scanning/Reconnaissance
4. Brute Force/Exploitation
5. Malware (Trojans, Worms)
6. Data Exfiltration

- IoT datasets

1. Web Attacks (SQL Injection, XSS)
2. DDoS/DoS
3. Malware (Ransomware, Spyware)
4. Password Attacks
5. DNS-based Attacks
6. Memory-based Exploits

## 4 Taxonomy of MFS-based IDS algorithms

This section presents our review methodology and then conducts a study of 36 recently presented IDS that use metaheuristic feature selection models and presents a taxonomy that highlights them in Fig. 3. It consists of SI algorithms, evolutionary algorithms, physical-based algorithms, human-behavior-inspired approaches, and novel hybrid approaches. These are organized into Tables 7, 8, 9, 10, and 11, respectively.

### 4.1 Review methodology

We followed an SLR methodology. To validate broad coverage of recent literature, Google Scholar was queried with the string:

(“Metaheuristic” AND “Feature Selection” AND “Intrusion Detection” AND “IIoT” OR “IoT”).

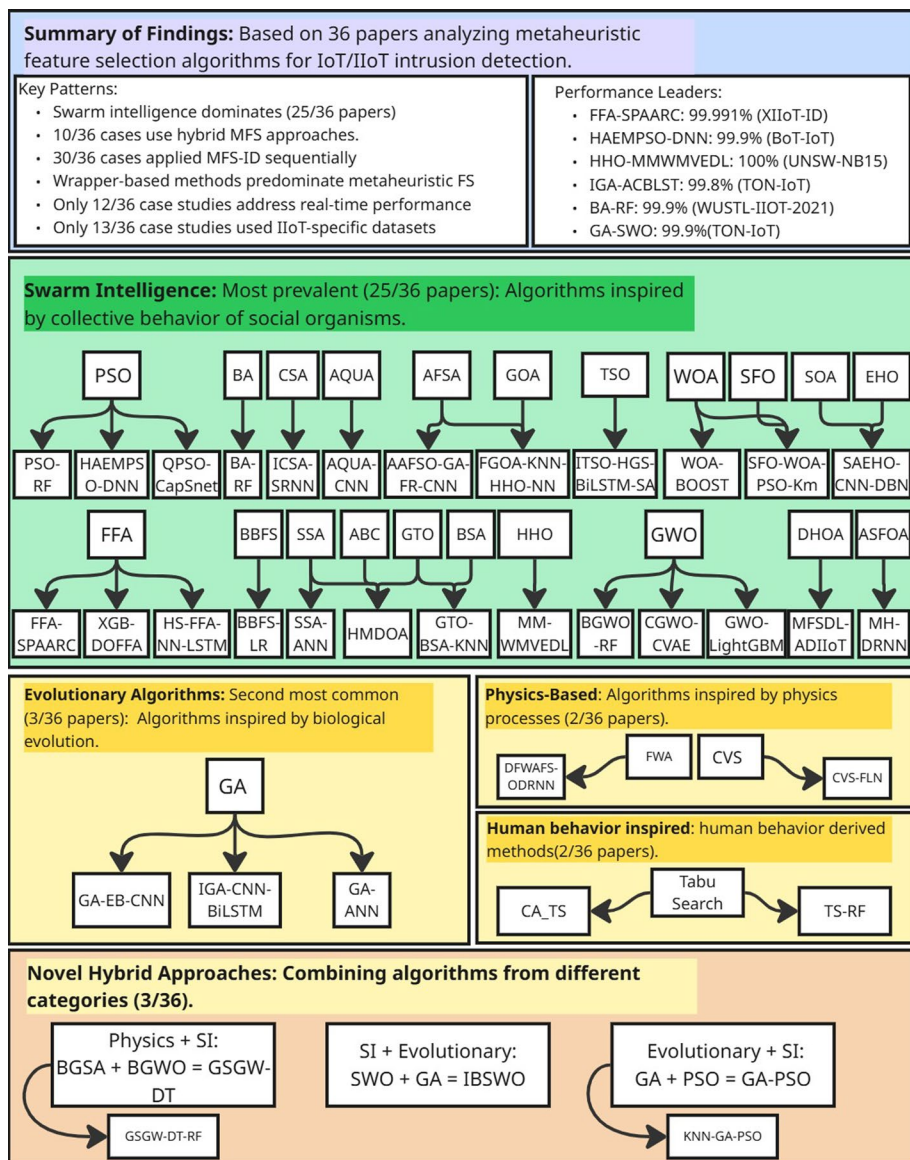


Fig. 3 Taxonomy of MFS-based intrusion detection algorithms for IIoT

**Table 7** Swarm intelligence metaheuristic algorithms for feature selection in IIoT intrusion detection

| Main topic                                                                        | Environment         | Feature selection algorithm | Intrusion detection algorithm            | Dataset(s) used                           | Accuracy                   |
|-----------------------------------------------------------------------------------|---------------------|-----------------------------|------------------------------------------|-------------------------------------------|----------------------------|
| Metaheuristics for FS with ML for malicious packet detection in computer networks | General Networks    | RF-PSO                      | RF                                       | NSS-KDD, kddcupdata_10%                   | 99.5787% (aggregate)       |
| Novel hybrid autoencoder and modified PSO FS for Multiclass ID in IIoT            | IIoT                | HAE-M-PSO                   | DNN                                      | UNSW-NB15, BoT-IIoT                       | 94.62%, 98.06% (aggregate) |
| Novel DL with SI-FS for intelligent ID in IIoT                                    | IIoT                | AAFSO                       | GA-FR-CNN                                | UNSW-NB15, BoT-IIoT                       | 94.488%, 93.756%           |
| IIoT ID Method Using ML and Optimization Techniques                               | IIoT                | BA                          | Random Forest                            | WUSTL-IIoT-2021                           | 99.9%                      |
| Enhancing IIoT IDS using hybrid DL-metaheuristics approach                        | IIoT                | HHO                         | RNN                                      | IoT-23, CICIDS2017, UNSW-NB15             | 98.12%, 99.98%, 100%       |
| Designing MFS for a DL model for a secure IIoT environment                        | IIoT                | SAEHO                       | CNN-DBN                                  | UNSW-NB15                                 | 92.8% (80% LR)             |
| ID for an IIoT data platform based on DL                                          | IIoT                | ASFOA                       | Deep Random NN                           | UNSW-NB15                                 | 99.2%                      |
| Hybrid meta-heuristics and XGBoost for IDS in IIoT                                | IIoT                | WOA                         | XGBoost                                  | IoTID20, UNSW-NB15                        | 99.2%, 99.5%               |
| Metaheuristic FS with DL for ID in the IIoT environment                           | IIoT                | DHOA                        | Cascade RNN                              | UNSW-NB15, UCI-SECOM                      | 99.31%, 97.88%             |
| Double-tier technique to prevent adversarial threats against IIoT networks        | IIoT (Industry 4.0) | Coati-GWO                   | Conditional Variational AE (CVAE)        | RT-IIoT2022                               | 99.91%                     |
| Comprehensive approach to ID in IIoT ecosystems                                   | IIoT                | Quantum-PSO_ANFIS           | CapsNets(Capsule Network), Augmented RNN | BOT-IIoT, TON-IIoT                        | 98.96%, 98.83%             |
| IDS for IIoT ecosystems integrating salp SI with ANN                              | IIoT                | Salp Swarm Algorithm (SSA)  | Artificial Neural Network (ANN)          | Edge-IIoTset, WUSTL-IIoT-2021, IoTID20    | 88.241%, 93.610%, 97.698%  |
| FFA-SPAARC hybrid technique for MFS-ID in Software Defined IIoT                   | IIoT                | FFA                         | SPARRC                                   | DDoS-SDN, XIIoT-ID                        | 99.9962%, 99.991%          |
| MFS-ID for botnet cyberthreats in IIoT                                            | IIoT                | Hybrid SFO-WOA              | PSO-KMeans                               | BOT-IIoT, UNSW-NB15                       | 99.8%, 99.5%               |
| DL-ID in ICS using Bio MFS in IIoT                                                | IIoT                | Improved CSA                | Stacking RNN                             | NSL-KDD                                   | 98.3%                      |
| Reliable ML model for Botnet detection in IIoT                                    | IIoT                | F-GOA-KNN                   | Improved HHO-ANN                         | N-Balot                                   | 98.7%                      |
| FS model using hybrid metaheuristic algorithms for IIoT ID                        | IIoT                | GTO-BSA                     | —                                        | NSL-KDD, UNSW-NB15, CICIDS-2017, Bot-IIoT | 95%, 98%, 71%, 94%         |



**Table 7** (continued)

| Main topic                                                                     | Environment  | Feature selection algorithm                          | Intrusion detection algorithm | Dataset(s) used                          | Accuracy                           |
|--------------------------------------------------------------------------------|--------------|------------------------------------------------------|-------------------------------|------------------------------------------|------------------------------------|
| ML tuning by diversity-oriented metaheuristics for Industry 4.0                | Industry 4.0 | DOFFA                                                | XGBoost                       | DTM, EFPADD, BoT-IoT, UNSW-NB15          | 100%, 100%, 99.52%, 94.94%         |
| ID system based on NN-LSTM merged with a hybridized statistical and FFA method | IoT          | Hybridized statistical and Firefly Algorithm (FFA)   | NN-LSTM                       | NF-BoT-IoT-V2, IoTID20                   | 98.42%, 89.54%                     |
| Advanced FS IoT and metaheuristic DL and AQO for IDS in IoT                    | IoT          | AQUO                                                 | CNN                           | NSL-KDD, CIC2017, KDD99, BoT-IoT         | 76.002%, 99.911%, 99.919%, 98.926% |
| IIDS using DL and Metaheuristic algorithms for IoT                             | IoT          | Improved tuna swarm optimization (ITSO)              | BiLSTM with SA                | Ton-IoT, Edge-IIoT                       | 99.44%, 99.37%                     |
| Ensemble MFS-NIDS using GWO and Lightgbm                                       | IoT          | GWO                                                  | Lightgbm                      | UNSW-NB15, CIRA_CIC_DoHBrw, and BoT-IoT  | 99.99%, 99.99%, and 99.84%         |
| FS-NIDS IoT using hybrid HMDOA and CNN classifier                              | IoT          | HMDOA(GTO, Artificial Bee Colony(ABC), SSA, and KNN) | CNN                           | NSL-KDD, CIC_MalMem_2022, and RT_IOT2022 | 96.54%, 79.09%, and 99.40%         |
| Multi-level IIoT IDS using BBFS and LR                                         | IIoT         | Bowebird courtship feature selection(BBFS)           | Logistic Regression(LR)       | WUSTL-IIoT                               | 99.70% and 100%                    |

**Table 8** Evolutionary algorithms for feature selection in IIoT intrusion detection

| Main topic                                                                              | Environment | Feature selection (FS) algorithm | Intrusion detection algorithm                | Dataset(s) used          | Accuracy                |
|-----------------------------------------------------------------------------------------|-------------|----------------------------------|----------------------------------------------|--------------------------|-------------------------|
| FS based on a GA for ID of ICS                                                          | ICS         | BGA                              | ANN                                          | SWAT, WADI               | 98.08%, 91.874%         |
| IDS for IoT environments utilizing an improved GA for MFS and a hybridized ACBLST model | IoT         | Improved Genetic Algorithm (IGA) | CNN-BiLSTM with Attention Mechanism (ACBLST) | TON_IoT                  | 99.8%                   |
| GA-FS and ensemble based CNN for ID in Smart IoT                                        | Smart IoT   | Genetic Algorithm                | Ensemble-bagging CNN                         | CIC-IoT-2023 , UNSW-NB15 | 99.73% , 95.33%         |
| Adaptive hybrid FS-ID framework for IIoT utilizing ANN and GA                           | IIoT        | Genetic Algorithm                | ANN                                          | IoTID20 , IoT-Botnet     | 99.7% (Merged datasets) |

**Table 9** Physics based algorithms for feature selection in IIoT intrusion detection

| Main topic                                                                   | Environment | Feature selection (FS) algorithm | Intrusion detection algorithm | Dataset(s) used       | Accuracy       |
|------------------------------------------------------------------------------|-------------|----------------------------------|-------------------------------|-----------------------|----------------|
| IDS using metaheuristic fireworks optimization-FS with DL on IoT environment | IoT         | FWA                              | DRNN                          | —                     | 96.11%         |
| A novel IoT-IDS based on meta-heuristic FS and classification                | IoT         | CVS                              | FLN                           | CI-CIDS-2017, BoT-IoT | 99.77%, 99.68% |

**Table 10** Human behaviour based algorithms for feature selection in IIoT intrusion detection

| Main topic                                          | Environment | Feature selection (FS) algorithm | Intrusion detection algorithm | Dataset(s) used | Accuracy (%) |
|-----------------------------------------------------|-------------|----------------------------------|-------------------------------|-----------------|--------------|
| Novel combinatorial optimization FS method for NIDS | General     | Tabu Search                      | Random Forest                 | UNSW-NB15       | 85.12        |
| Novel combinatorial FS algorithms for network ID    | IoT         | CA-TS                            | Random Forest                 | TON-IoT         | 99.50        |

The string gave 1050 results; we screened the first 100 search results ranked by relevance for inclusion in our work. We filtered these results based on the following inclusion and exclusion criteria:

#### 4.1.1 Inclusion

- Studies published in peer-reviewed articles and conferences of the 5th edition and above.
- Studies exclusively using metaheuristics for feature selection for intrusion detection in an IIoT-relevant context.

**Table 11** Novel hybrid approaches for feature selection in IIoT intrusion detection

| Main topic                                                    | Environment | Feature selection (FS) algorithm | Intrusion detection algorithm | Dataset(s) used               | Accuracy            |
|---------------------------------------------------------------|-------------|----------------------------------|-------------------------------|-------------------------------|---------------------|
| Metaheuristic ensemble FS for ID in IoT-enabled networks      | IoT         | GSGW-DT                          | Random Forest                 | UNSW-NB15                     | 99.41%              |
| Adjusted BSWO algorithm for ID for IoT                        | IIoT        | IBSWO(GA-SWO)                    | Unspecified                   | UNSW-NB15, TON-IoT, NCTLR-IoT | 98.7%, 99.9%, 99.7% |
| Autonomous vehicle metaheuristic optimization for IDS in IIoT | IIoT        | GSA-PSO                          | XGBoost                       | CAN-dataset                   | 79.11%              |

- Publications available in English
- Studies published between 2018 and 2025
- Studies published by reputable publishers such as Springer, Wiley, MDPI, IEEE, Elsevier, and Sage.
- Studies that present empirical results of our relevant context.

#### 4.1.2 Exclusion

- Non-peer-reviewed articles (e.g., blogs, opinions, magazines)
- Studies that use metaheuristics for purposes other than feature selection for ID-IIoT.
- Publications prior to 2018
- Studies published between 2018 and 2025
- Research papers published in journals below Q3.
- Studies that present empirical results of our relevant context.
- Duplicate studies or extended versions of previously published work without significant new contributions.

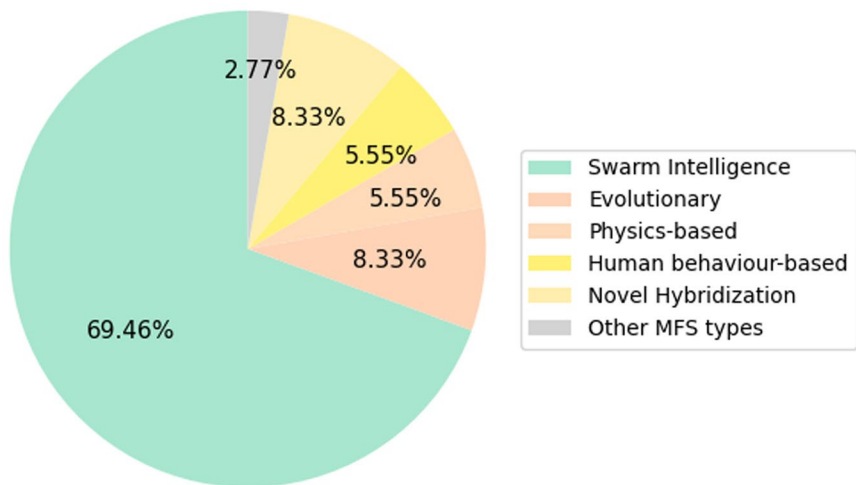
We also added some case studies manually or through reviewers' recommendations.

#### 4.2 Taxonomy

This subsection presents the taxonomy and a discussion about our findings in this review. Figure 3 visualizes a taxonomy of the FS metaheuristic algorithms, highlighting the algorithms and hybridizations with the highest performance, their numbers by type and the unique hybridizations that were presented as novel methodologies.

The taxonomy displayed significant patterns in how MFS algorithms are developing for IoT/IIoT IDS. The SI technique dominates the landscape (69.44%), with variants of PSO, GWO, and FFA being the most used, and cases of using PSO, FFA, BA, and HHO showing strong performances. This prevalence is likely due to the natural capabilities of swarm algorithms to equalize between exploration and exploitation in high-dimensional attribute spaces typical to networking traffic data. The distribution usage of MFS algorithms across MFS types can be seen in Fig. 4, which shows the MFS usage per category.

## MFS-based algorithms for ID in IIoT by origin



**Fig. 4** MFS algorithms usage per metaheuristics category

Future metaheuristic feature selection research should prioritize:

- Standardizing computational efficiency metrics while also reporting accuracy is essential.
- Validation on resource-constrained hardware reflecting actual deployment conditions
- Industrial-specific dataset utilization with domain-relevant attack vectors
- Dynamic feature selection approaches that can adapt to evolving attack patterns
- Explainability Mechanisms to Support Security Analyst Decision Making

The most promising direction seems to involve developing adaptable hybrid techniques that can dynamically balance accuracy and computational requirements based on the threat context and available resources, effectively optimizing the optimization process itself.

### 4.3 Swarm intelligence MFS

This subsection summarizes case studies of SI-based MFS algorithms.

Shanbhag et al. (2024) presented Leveraging Metaheuristics for FS with ML Classification for Malicious Packet Detection in Computer Networks. This research is meant to advance the use of nature-inspired FS algorithms in IDS. They experimented with the NSS-KDD AND kddcupdata\_10% dataset using 48 combinations of the FS and ML algorithms. They ended up proposing three things: Genetic Algorithm (GA), Gray Wolf Optimization

(GWO), and Particle Swarm Optimization (PSO). All were combined with one of the four ML algorithms: Random Forest (RF), Logistic Regression (LR), Gaussian Naive Bayes (GNB), and Random Forest (RF). The combination achieving the highest was RF-PSO\_RF with an accuracy of 99.5787%; second was RF-PSO\_DT with 99.5019%, and third was RF-GA\_GNB with 99.4150%. They also excelled in recall, F-measure, and precision. One notable missing piece of information in this paper is the computational complexity of hybrid algorithms.

Saheed et al. (2023) presented a novel hybrid autoencoder and modified particle swarm optimization feature selection for intrusion detection on the Internet of Things network. They proposed HAEMPSO, a hybrid autoencoder, and modified the PSO attribute selection algorithm, followed by a DNN for classification. PSO is a swarm intelligence FS technique used to solve complex optimization problems. Experiments were conducted using 10-fold cross-validation on two datasets, UNSW-NB15 and BoT-IoT. HAEMPSO-DNN gave accuracies of 98.8% and 99.9% on the generic attack and benign category, while it had 99.22% and 97.54% on the DDoS HTTP and benign category. It also had a detection rate of 99.9%, 99.7%, 99.79%, and 99.92% in the same categories, respectively. In the BoT-IoT dataset, the DNN model scored 97.54%, 98.22%, 98.50%, and 99.22% on accuracy for Benign, DDoS UDP, DDoS TCP, and DDoS HTTP, respectively. Moreover, it got 99.22%, 97.21%, 97.72%, and 97.79% for detection rate in the same categories. However, this research limits the DL testing to DNN and does not show results of what would happen if HAEMPSO were run on different DL models such as CNN or RNN. Furthermore, it does not provide comprehensive clarification of the time complexity.

Anushiya and Lavanya (2023) presented a new DL with swarm-based FS for intelligent ID for the IoT. They proposed two new algorithms: GA\_FR\_CNN for intrusion detection and AAFSO for attribute selection. Where GA is the natural selection-inspired genetic algorithm, FRCNN stands for fast recurrent convolutional neural networks. AAFSO, for assimilated artificial fish swarm optimization, is a nature-inspired FS algorithm based on random search by simulating the movements of schools of fish and the intelligence that creates it. The test was carried out on the basis of the UNSW-NB 15 and BOT-IoT datasets, where AAFSO\_GA-FR-CNN scored an accuracy of 94.488% and 93.7756% and an error rate of 5.5120% and 6.2244%, respectively. It also excelled in precision, recall, and F measures, achieving results similar to the accuracy in both datasets. Despite the two newly proposed algorithms and the results they achieved, the paper lacks any description of their computational complexity.

Gaber et al. (2023) presented an Intrusion Detection Method Using the IIoT.Machine Learning and Optimization Techniques. They proposed an IDS for IIoT that consists of two nature-inspired algorithms, PSO and the bat algorithm (BA), and then a random forest for classifying malicious traffic. The tests were carried out on the industrial WUSTL-IIOT-2021 data set using standard evaluation metrics. The combination of PSO-RF achieved an accuracy of 99.7%, an F-score of 95.6%, a precision of 95.4%, and a recall of 95.8%. On the other hand, BA-RF landed at 99.9%, 99.6%, 99.6%, and 99.6% in the same measures, respectively. The computational complexity of PSO-RF and BA-RF is said to be small since the parameters have been highly reduced. A notable advantage of this paper is the usage of a specifically designated IIoT dataset. However, it could have benefited from using a second, more general dataset such as TON\_IoT.

Sanju (2023) presented Enhancing intrusion detection in IoT systems: A hybrid meta-heuristics-deep learning approach with an ensemble of recurrent neural networks. An SI algorithm called Harris hawk optimization (HHO) is used for FS, and RNN is used as the classifying technique. The open access datasets IoT-23, CICIDS2017, and UNSW-NB15 were used for experiments. The final proposed model, called MM-WMVEDL, yielded accuracies of 98.12% and 99.98%, an F1-score of 98.18% and 99.99%, precisions of 98.06% and 99.99%, and a recall of 98.31% and 99.98% on IoT-23 and CICIDS-2017, respectively. Furthermore, in the UNSW-NB15 dataset, the model obtained a perfect 100% score on all measurements. However, the time complexity of this model was not discussed in this paper, which is a concern given the complexity of RNN.

Sagu et al. (2023) presented the design of metaheuristic optimization algorithms for a DL model for a secure IoT environment. They proposed seagull-adapted elephant herding (SAEHO) and self-updated mouse and cat (SU-CMO) as new animal behavior-based feature selection for DL. Then, they proposed their usage in two merged DL classing algorithms, where SAEHO was hybridized with CNN-DBN and SU-CMO was hybridized with Bi-LSTM-GRU. These hybridized models were executed and evaluated on a major dataset in this field, UNSW-NB15. They used three different learning rate percentages. At a 60% learning rate, SAEHO-CNN-DBN had 90.5% accuracy and a 77.6% F-measure. While using 70% and 80% learning rates, it achieved 90.8%, 79.0%, 91.6%, and 75.0% on the same measures, respectively. This research has a technical problem; it used a second dataset but only referred to it as “dataset 2,” and the link provided to it has expired. Furthermore, the research did not detail the time complexity of its proposed algorithms. Besides this, the paper acknowledges that this solution is considered good but not necessarily optimal.

Li et al. (2024) presented an anomaly detection algorithm for an IIoT data platform based on DL. The adaptive sunflower optimizing technique (ASFOA) is an FS nature-inspired algorithm that follows the movement of sunflowers. This paper combines this technique with a deep random NN model to produce the novel MH-DRNN IIoT IDS. The novel IDS was executed and evaluated on the UNSW-NB15 dataset. Taking into account five different learning rates, MH-DRNN received an overall accuracy of 99.2%, the averages of other measures being 98.456% for recall, 98.44% for f score and 98.466% for precision. This paper could have benefitted from using a secondary dataset to further solidify the general capability of the proposed model. Furthermore, it does not include a detailed time complexity description.

Bajpai et al. (2024) presented a hybrid metaheuristics and XGBoost approach for IDS in IoT. This paper proposed an IDS integrating XGBoost as its core ID technique, with the whale optimization algorithm (WOA) preceding it. WOA is used as FS. WOA is inspired by the social behavior of humpback whales during their hunt. It was chosen to perform here due to its simplicity. parameters. IoTID20 and UNSWNB15 datasets were used for the execution and evaluation of WOA-BOOST. The WOA-BOOST model landed at an accuracy of 99.2% and 99.5%, an AUC of 98.91 and 99.01, a recall of 99.91% and 99.00%, a precision of 98.67% and 99.05%, and an F1 score of 99.30% and 99.03% on the two datasets, respectively. This paper achieved great results in optimizing an algorithm through metaheuristic FS. However, it lacks details on the computational complexity of this algorithm, which would be a concern in any optimization of an algorithm involved.

Chander and Upendra Kumar (2023) presented an MFS with DL, which enabled cascaded RNN for anomaly detection in the IIoT environment. The deer hunting optimizing

technique (DHOA) was proposed as a novel FS algorithm; it is proposed by the hunting methods of deer by humans. Cascaded RNN was incorporated as the IIoT classifier. The sparrow search optimizer (SSO) was used to optimize the CRNN model, enhancing its DR ability. They used UNSW-NB15 and UCI-SECOM for the execution and evaluation of the proposed model. The complexity of DHOA is not mentioned, while it can be inferred that the complexity of CRNN is said to depend on the number of layers and how many nodes are in each layer. In the two data sets, the final proposed MFSDL-ADIIoT framework scored an average accuracy of 99.31% and 97.88%, respectively. It also had a precision of 67.09% and 92.42%, a recall of 60.33% and 89.87%, a specificity of 99.58% and 89.87%, and an F-measure of 60.35% and 91.10%, respectively. The paper lacks a detailed exploration of computation complexity, which is concerning, given that it used RNNs. The algorithm has struggled with precision and recall on UNSW-NB15, which could indicate problems in identifying true positives and detecting intruders efficiently. Furthermore, the low F1 score on the same dataset indicates that the algorithm may lack performance when balancing false negatives/positives. Table 1 summarizes all the above FS algorithms.

Prasad et al. (2025) suggested a double-tier technique to prevent adversarial threats against IoT networks. This research merged FS with advanced identification methods. It aims to improve security in Industry 4.0 ecologies. The initial tier includes FS with the Coati-GWO, and the later tier utilizes a conditional variational AE (CVAE) with optimized parameters for threat recognition. The RT-IoT2022 dataset was utilized for evaluation, where the proposed model scored an accuracy of 99.91% and an F-score of 98.11%. This research still lacks real-world experimenting, and the model is solely tested on a single dataset, which creates scalability and generalizability uncertainties. Furthermore, this paper lacks detailed exploration of time complexity.

Logeswari et al. (2025) suggested a comprehensive approach to ID in IoT ecosystems. This ID system applied the quantum-inspired PSO MFS algorithm, combined with a fuzzy logic algorithm short-named ANFIS. This two-tier optimization minimized features from 46 to 22 and from 45 to 20 in the BOT-IoT and TON-IoT datasets, respectively. The CapsNets DL architecture was hybridized with an Attention-Augmented RNN (ARNN) as a multi-stage classifier for both binary and multi-attack scenarios. The suggested model scored an overall accuracy of 98.96% in BOT\_IoT and 98.83% in TON\_IoT. However, this implementation still lacks real-world experimentation. Furthermore, the authors acknowledge that their hybridized model has high computational overhead compared to traditional FS.

Alzubi et al. (2025) suggested an IDS for IoT ecosystems that integrates salp swarm algorithm (SSA) with artificial NN (ANN). The authors utilized three datasets: Edge-IIoT-set, WUSTL-IIoT-2021, and IoTID20. Were it landed accuracies of 88.241%, 93.610%, and 97.698%, respectively. However, the authors acknowledge that the model may struggle with long training times on large datasets and the lack of real-world experimentation.

Thamer Francis et al. (2025) A hybrid firefly-based attribute selection and split-point mechanism (SPAARC) for securing Software Defined-IIoT. It is an enhanced decision tree (DT) technique that hybridizes node attribute sampling (NAS) and split point sampling (SPS). DDoS\_SDN and XIIoT\_ID were used as experimentation data, and the model scored an accuracy of 99.9962% and 99.991%, respectively. This paper, however, while it states that SPAARC is 70% faster than the classic CART DT, does not explicitly mention its time complexity. Furthermore, the model had not been tested in real-world datasets.

Maazalahi and Hosseini (2025) proposed an ML and an MFS for botnet cyber threat detection in IoT. They proposed a novel hybrid SI-based MFS model between Sailfish Optimizer (SFO) and Whale Optimization Algorithm (WOA). They operate by SFO removing redundant features and defining the initial population, and then WOA refining the initial population to identify the optimum feature set. A hybrid PSO-Kmeans follows this mechanism, which is the model for ID. This hybridization was tested on BOT-IoT and UNSW-NB15 datasets, scoring accuracies of 99.8% and 99.5%, respectively. This work is one of the very few works that mention computational complexity in detail. It shows that the model is slower than basic metaheuristics that do not integrate clustering steps.

Kocherla et al. (2024) proposed a DL-based stacking RNN (SRNN) for ID in industrial control systems utilizing bio-derived MFS in IIoT. This study proposed the DLAD model, a bio-inspired metaheuristic technique derived from DL techniques for attack recognition. An improved crow search algorithm (ICSA) is extracted using the DLAD model and utilized as an MFS. The Harris Hawks Optimizer (HHO) is functionalized to get the optimum tuning of the SRNN model's parameters. The suggested model was tested on the NSL-KDD dataset, scoring an efficiency of 98.3%, defeating other recently proposed models. This work, however, focused on performance accuracy and does not clarify what the time complexity of the proposed model was.

Taher et al. (2023) proposed a reliable ML model for IIoT botnet detection with an MFS. A triple feature hybrid artificial fisher swarm algorithm (AFSA), grasshopper optimization (GOA), and KNN algorithms were used, with fisher and GOA being swarm intelligence techniques. This technique is followed by a hybrid improved HHO and NN for threat detection. The models were tested on the N-BaIoT dataset, scoring an accuracy of 98.7% across multi-classification of 10 attack types. The researchers described the time complexity to have been significantly reduced, although they did not describe it formally. Furthermore, the testing of this algorithm is limited to a single dataset only, which questions how it will perform on datasets with different traits.

Kareem et al. (2022) presented an effective feature selection model using hybrid metaheuristic algorithms for IoT intrusion detection. They proposed a novel hybrid Gorilla Troops Optimizer for Bird Swarms (GTO-BSA) attribute selection algorithm. It is a nature-inspired algorithm in which BSA is used to increase GTO's ability to identify viable regions with the most suitable solutions. GTO-BSA is then followed by a KNN algorithm for ID and tested on various ID and IIoT datasets, including NSL-KDD, UNSW-NB15, CICIDS-2017, and Bot-IoT. GTO-BSA was compared to other nature-inspired algorithms such as the original GTO, BSA, HGS, and PSO. It achieved a higher convergence rate and higher quality assurance solutions with an accuracy of 95% in NSL-KDD, 98% in CICIDS-2017, 71% in UNSW-NB15 and 94% in the Bot-IoT datasets. GTO-BSA also excelled in other measures such as fitness, sensitivity, and specificity. GTO-BSA has a computation complexity of  $O(T*N*D)$ , where  $T$  is the highest possible number of iterations,  $D$  is the problem dimension, and  $N$  is the population size. In conclusion, GTO was upgraded through a hybrid swarm intelligence algorithm called BSA; the study, however, lacks real-world deployment and testing of this combined algorithm.

Jovanovic et al. (2024) presented metaheuristic-based feature selection and ML tuning using diversity-oriented firefly metaheuristics for Industry 4.0. DOFFA is a modified version of the firefly technique (FFA); FFA is an SI nature-inspired FS algorithm that simulates a firefly's movement. It is also used for metaheuristic hyperparameter optimization



(MHO). DOFFA is combined with an extreme learning machine (ELM) or XGBoost to classify solutions like IDS. Distributed transformer monitoring (DTM), elderly fall prediction and detection (EFPADD), BoT-IoT, and UNSW-NB15 are utilized as experimental datasets. The computational complexity of DOFFA is said to be the same as FFA with  $O(T \times N^2)$ . ELM-DOFFA scored a perfect accuracy of 100% on both the DTM and EFPADD datasets, while XGB-DOFFA excelled on the other two, with 99.52% in BoT-IoT and 94.94% in UNSW-NB15. On the BoT-IoT, ELM-DOFFA had the same weighted average precision, recall, and f-measure of 90.2%, while XGBoost had 99.5%. On the UNSW-NB15, ELM-DOFFA had 90.2%, 91.9%, and 89.9%, and XGB-DOFFA had 94.9%, 94.9%, and 94.3% of average weight using the same metrics. This paper included everything; it does not lack any information, which is excellent. However, I would criticize this paper's length since it consists of 33 pages, which is relatively long.

Ogunseyi and Thiyagarajan (2025) suggested an ID system based on ANN-LSTM merged with a hybridized statistical and FFA (HS-FFA) method for MFS. The model reduces useless features and improves the accuracies to 98.42% on the NF-Bot-IoT-V2 dataset and 89.54% on the IoTID20 dataset. Basic MFS also decreases computational overhead and identifies key features that greatly contribute to threat detection. This is one of the very few papers in this field that details time complexity, as

It is said to be  $O(n \times k + k^2)$ . This research has implicit IIoT relevance, as it directs the model for usage towards scalability for large-scale networks and quasi-real-time operation capabilities (8.5 ms per sample).

The proposed technique, however, is yet to be experimented with in a real-world environment to see how the quasi-real-time performs. Furthermore, the model appears to struggle to detect zero-day attacks, as it had a low 54.76% accuracy in that particular class.

Fatani et al. (2021) presented an advanced feature extraction and selection approach using

deep learning and the Aquila optimizer for IoT IDS. AQUO is a population-based SI technique commonly used for feature selection. This paper combined a binary version of AQUO with the CNN classing method. NSL-KDD, CICIDS2017, KDD99, and BoT-IoT were used as execution and evaluation datasets. The model got testing average accuracies of 76.002%, 99.911%, 99.919%, and 98.926%, respectively, becoming the Optimus performer on all four datasets except BoT-IoT. It also had the same results with average F-measure, precision, and sensitivity. Despite high accuracy, this paper lacks a formal description of the resulting algorithms' time complexity, which is crucial in the usage of any neural network model.

Ablehai (2025) proposed an AI-driven cybersecurity IIDS using DL and metaheuristic algorithms for IIoT. An improved tuna swarm optimization (ITSO) technique was used for MFS, followed by an HGS hyperparameterized ML model BiLSTM with self-attention. Ton-IoT and Edge-IIoT datasets were capitalized for evaluation. The proposed model scored an accuracy of 99.44% and 99.37%, respectively. The presented hybridization, however, despite its robustness, still lacks real-world experimentation. Furthermore, the computational overhead isn't fairly clarified in the model.

Mahato and Dutta (2025) proposed a GWO and LightGBM FS-IDS for detecting threats on IoT. The model was executed sequentially on the UNSW-NB15, CIRA\_CIC\_DoHBrw, and BoT-IoT and scored accuracies of 99.99%, 99.99%, and 99.84% with execution times of 27.737 s, 44.66 s, and 1.40 s, respectively. Thus, the model is said to be lightweight.

The Bot-IoT dataset is highly relevant to IIoT, and thus, this research was included as a case study. This research, while mentioning preventing overfitting in LightGBM through max depth, however, lacks detailed analysis of the overfitting problem, and the model still requires further testing in real-time environments.

Gong et al. (2025) presented HMDOA, a hybrid metaheuristic dynamic optimization model. The model was built with a multi-mechanism PB-based metaheuristic, functionalizing GTO for initial population, Artificial Bee Colony (ABC) for local search, SSA for global exploration, the Levy flight mathematical mechanism for randomizing perturbations, and finally KNN as a fitness evaluator. This FS model is followed sequentially by a CNN classifier, which scored accuracies of 96.54%, 79.09%, and 99.40% on the NSL-KDD, CIC\_MalMem\_2022, and RT\_IOT2022 datasets, respectively. The time complexity of the model is addressed to be  $O(T \cdot N \cdot d)$ , while space complexity is  $O(N \cdot d)$ . This research, however, lacks proper overfitting analysis of the proposed algorithm, which is concerning given the high accuracies reported on NSL-KDD and RT-IoT2022. This research is strongly relevant to IIoT through the RT-IoT2022 dataset, as it is built of real-time and IIoT network traffic. Thus, we included it as a case study.

Mallidi and Ramisetty (2025) proposed a multi-layer IDS for IIoT utilizing the Bowebird courtship feature selection (BBFS) algorithm. This FS model was executed in two modes: in a wrapper approach followed sequentially by a logistic regression (LR) model in the edge tier and in a filter approach followed sequentially by a 1D CNN in the cloud tier, with both tiers included in the framework. It was tested on the WUSTL-IIoT dataset and scored accuracies of 99.70% and 100% on the aforementioned layers, respectively. The authors detail the time complexity of BBFS as  $O(T \times N \times (\log N + |F| \times C(F)))$ , thus deducing its lightweightness and suitability for real time, with Edge LR inference minimized from 52 ms to 38 ms and cloud CNN minimized from 27.86 s to 18.28 s. Overfitting avoidance is addressed on the filter-based BBFS-CNN model; however, no overfitting is addressed on the metaheuristic wrapper-based BBFS-LR method, which is a notable weakness of this paper.

#### 4.4 Evolutionary

This subsection presents case studies that utilized evolutionary-based MFS algorithms. A summary of them can be seen in Table 8.

Fang et al. (2024) presented an FS based on a GA for ID of industrial control systems (ICS). Binary genetic algorithm (BGA) is a nature-inspired algorithm based on natural selection commonly used for FS. This paper has adjusted it to suit the real-time necessity of ICS features. The research also identified artificial neural networks (ANN) as the ideal ID technique for ICS. Experiments used real-world ICS datasets extracted out of the SWaT and WADI testbeds. Using 5-fold cross-validation, the NN classifier defeated all other ML algorithms with an accuracy of 98.08% and 91.874% and a response time of 87.13s and 16.97722s, respectively. Despite the elevated records, it is suggested that multi-objective GAs have the potential to get even higher scores if explored efficiently, which is something this research does not include. Furthermore, the research does not provide detailed analyses of the time complexity.

Fang et al. (2025) also presented an IDS for IoT environments utilizing an improved GA (IGA) for MFS and a hybridized ACBLST model, that is, a CNN-BiLSTM with the Attention Mechanism (AM). This was evaluated on the TON\_IoT dataset, scoring a finalized

accuracy of 99.8%. This research is concerning in terms of generalizability, as it relied only on a single dataset in the research. Furthermore, while the authors acknowledge the model would struggle with heterogeneous and real-time applications data, it does not include any formal time complexity analysis with mathematical notation.

Sharma and Singh (2025) proposed a GA-based FS and CNN-based ensemble model for ID in smart IoT environments. GA was utilized to identify the most relevant features from the network packets, while a CNN-based bagging model was hybridized with three pre-trained CNN architectures (Xception, VGG16, and VGG19) under Random Hyperparameter Optimization (RHO). The models were tested on CIC-IoT-2023 and UNSW-NB15, scoring accuracies of 99.73% and 95.33%, respectively. Despite the good results, this research lacked a description of the time complexity of the proposed model.

Khan et al. (2025) proposed an adaptive hybrid framework for ID in IIoT using GA for MFS and ANN with L2 regularization and dropout for ID. This model was evaluated on the IoTID20 and IoT-Botnet datasets, scoring an accuracy of 99.7%. The framework is presented to be applicable in real-time IIoT security monitoring and ID. However, this research does not detail the time complexity of this model and has some misclassifications in low-profile attack categories such as probe, U2R, and R2L.

#### **4.5 Physics based**

This subsection presents case studies that utilized physics-based MFS algorithms. A summary of them can be seen in Table 9.

Jayasankar et al. (2024) proposed an IDS using metaheuristic fireworks optimization-based feature selection with deep learning on an Internet of Things environment. They proposed a combination of the fireworks FS algorithm (FWA) and deep recurrent neural network (DRNN)-based ID to create the DFWAFS-ODRNN IDS. FWA is a swarm intelligence technique inspired by fireworks sparks. The model was executed and evaluated on a benchmark IDS dataset, resulting in a high accuracy of 96.11%, a DR of 97.31%, and an FPR of 3.03%. The authors, however, do not explicitly mention nor reference the dataset's name and link nor explain the contents or the features included in the dataset, which is a significant flaw of this paper. Knowing the dataset type could also indicate the need for a secondary dataset.

Geetha et al. (2024) presented CVS-FLN, a novel IoT-IDS model based on metaheuristic FS and the NN classification model. CVS, which stands for chaotic vortex search, is a physics-inspired metaheuristic FS mechanism built on chaotic maps and vertex-based search. This paper combines CVS with a fast learning network (FLN), a version of ANN, to create the CVS-FLN anomaly detection system. CVS-FLN was tested on two publicly accessed datasets, CICIDS-2017 and BoT-IoT. The complexity of the FLN model is not mentioned; however, it is emphasized that it has been reduced by setting weights to all hidden layers during training and minimizing error by using a linear output layer methodology. CVS-FLN got an accuracy of 99.77% and 99.68%, a DR of 99.81% and 99.11%, a specificity of 99.92% and 99.83%, a precision of 99.60% and 99.30%, and an F measure of 99.72% and 99.21% in the two mentioned datasets, respectively. This research was able to achieve very high results. However, it does not go into the details of the time complexity of each algorithm. Additionally, it lacks real-world deployment, execution, and evaluation.

## 4.6 Human behaviour inspired

This subsection presents case studies that utilized human behavior-based MFS algorithms. A summary of them can be seen in Table 10.

Nazir et al. (2023) presented a novel feature selection algorithm in IoT networks for intrusion detection. They proposed a new wrapper-based hybrid FS algorithm built on the Cellular Automata (CA) engine, the nature-inspired Tabu Search (TS), and Random Forest (RF). CA-TS operates as a search algorithm, while RF is the predictor. They conducted experiments on the TON-IoT dataset, and the combination of RF and CAT-S excelled over the other combinations, such as CNN and the correlation coefficient, scoring an accuracy of 99.50% and a false positive rate of 0.004%, using only 13 features. The cost of this function is said to be 4.496. However, this algorithm was only tested on a single data set and is yet to be tested on other popular data sets like CICIDS-2017 and Bot-IoT. In addition, it lacks real-time deployment considerations.

Nazir and Khan (2021) presented a novel combinatorial optimization-based feature selection method for network intrusion detection. This paper utilized a wrapper-based FS algorithm named tabu search (TS) and the random forest (RF) classifier for threat recognition. TS is a human behavior-inspired algorithm, as it simulates human memory and adaptive decision-making. TS is a general metaheuristic technique based on evaluating neighbor nodes created in a random subset of features, placing the newly visited nodes in a tabu list, and iterating until the last iteration ends. The hybrid TS-RF and other ML algorithms were executed and evaluated on the UNSW-NB15 dataset. TS-RF scored an accuracy of 83.12% and an FPR of 3.7% considering 16 attributes. Notable weaknesses here include the usage of a single dataset, which could have benefited from a secondary datasetlike TON\_IoT, and, moreover, the class imbalance mentioned in UNSW-NB15 itself.

## 4.7 Novel hybridizations

This subsection presents case studies that hybridized and used novel metaheuristics of different types. A summary of them can be seen in Table 11.

Dey et al. (2023) presented A metaheuristic-based ensemble FS framework for cyber threat detection in IoT-enabled networks. They proposed an attack detection framework that uses nature-inspired ensemble attribute selection using two binary (equipped with a decision tree) algorithms called the gravitational search algorithm (BGSA) and Gray Wolf optimization (BGWO), hybridizing them into GSGW-DT. GSGW-DT was followed by a classical method based on ensemble learning, such as random forest, to evaluate network traffic. This hybridization is unique because it is combining a physics-inspired algorithm with an SI-inspired algorithm. This hybrid FS-ID framework was tested in the UNSW-NB15 data set and achieved a near-perfect accuracy of 99.41%, detection rate (99.09%), F1 score (99.33%) and lowest false positive rate (0.03%). This paper mainly uses wrapper-based FS, as it usually obtains higher classification accuracy. One notable weakness of this paper is that it was tested only on a single dataset, UNSW-NB15, and this is a general network attack detection dataset and not specific to IoT or IIoT traffic. Furthermore, it had not detailed the time complexity of its proposed model.

Mohammad Shtayat et al. (2025) presented an Improved Binary Spider Wasp Optimization (IBSWO) Algorithm for ID for the IIoT, where IBSWO is utilized as the core FS algo-

rithm for this IDS. It consists of merging two nature-inspired algorithms, SWO and genetic algorithms. The GA's standard crossover was replaced with the flat crossover operator for enhanced evaluation. Three datasets were taken advantage of: UNSW-NB15, TON\_IoT, and NCTUK-IIOT, in which IBSWO scored accuracies of 98.7%, 99.9%, and 99.7%; a precision of 98.63%, 100%, and 99.7%; a recall of 98.4%, 99.7%, and 99.7%; and an F-measure of 98.5%, 99.7%, and 99.7% on the three previously mentioned datasets, respectively. While the proposed IBSWO is an FS algorithm designated for ID in IIoT, it lacks specific explanations and experiments on actual ID algorithms used to classify the data. Furthermore, it does not provide a formal time complexity description for the proposed models.

Dakic et al. (2024) presented ID using metaheuristic optimization within IIoT systems and autonomous vehicle software. They proposed two threat recognition classifiers, XGBoost and KNN, enhanced by a modified SI PSO algorithm for FS. Modifications on PSO included the inclusion of crossover and mutations as new genetic algorithm functions to generate the genetic self-adjusted PSO algorithm. This hybridization is for bettering exploration and exploitation and integrating quasi-reflexive learning (QRL) to improve the diversity of the first population. The KNN-GA-PSO and XG-GA-PSO were tested on the controller area network (CAN) dataset and compared to other ML algorithms. They landed accuracies of 79.10% and 79.11%, error rates of 0.208868% and 0.208868%, and Cohen's kappas of 0.076177% and 0.076166%, respectively. The weakness of this paper relies on the dataset; it is field-specific and experiments on a secondary, more general dataset. Despite the improvement of these models on the measurements, it appears accuracies can be improved. Furthermore, the paper does not clarify the time complexity of its presented model.

#### **4.8 Other metaheuristic types**

There is this single unique case in Quincozes et al. (2022), where they utilized GRASP, a stochastic single-solution-based optimization metaheuristic for FS. They presented an expanded evaluation of MFS for ID in the cyber-physical systems perception layer. GRASP is classified as a local search-based approach. They used the expert classifier approach for ID rather than a specific model, each focusing on detecting a specific type of cyber threat. The datasets utilized for evaluation were WSN-DS and the IIoT's SWaT dataset. Multi-class classification was applied to four attack types that were targeted. The attack types targeted were blackhole, flooding, grayhole, and TDMA. Those were targeted on WSN-DS, while binary classification was applied for injection attacks on SWaT. The proposed models employed the expert classifier approach for the multiclass scenario, with scoring accuracies ranging from 99.26% to 99.99%, and the random tree classifier, which achieved a score of 99.65% on the injection attacks. The study discovered that different attacks required different feature subsets for optimal detection. This paper, like many in this field, lacks a detailed clarification of the time complexity of the presented models.

### **5 Technical analysis and discussion**

This section presents a crucial technical evaluation of the revised methodologies, focusing on enriched FS techniques and ID headways in IIoT (IIoT) environments. We examine algorithmic curations, performance trade-offs, and unaddressed challenges.

## 5.1 Effective methodologies

The combination of PSO-RF emerged as a high-scoring approach (Shanbhag et al. 2024), as well as the PSO-DL methodology, appearing in two case studies. The PSO-RF pairing's popularity originates from PSO's direct implementation and RF's inherent robustness, continuously scoring a minimum of 99% accuracy on well-known datasets like NSL-KDD and CICIDS-2017. The PSO-DL techniques also had high scores, such as the HAEMPSO-DNN with 99.90% on the TON\_IoT dataset. However, PSO-RF risks overfitting to dataset-specific patterns, reducing generalization, while the PSO-DL technique faces scalability challenges in large-scale IIoT applications due to its high computational requirements. Likewise, The integration of GWO-RF and GWO-DL is as effective but struggles with similar vulnerabilities. Dey et al. (2023) and Prasad et al. (2025). There were many other usages of MFS + DL (e.g., Anushiya and Lavanya 2023; Fatani et al. 2021). Fang et al. (2024), and yes, all score high results but again struggle with the same computational requirements paradox.

Another common methodology is MFS + gradient boosting. In Jovanovic et al. (2024), Dakic et al. (2024), and Bajpai et al. (2024), XGBoost was proposed as the best performer, but in all cases it was preceded by an MFS, that is, FFA, GSA-PSO, and WOA, respectively. Also, GWO-LightGBM demonstrated strong performance, leveraging GWO's effective exploration-exploitation balance to enhance LightGBM's gradient boosting capabilities, as evidenced by Mahato and Dutta (2025), scoring 99.41% accuracy on UNSW-NB15.

Hybrid SI methods, including GTO-BSA (Kareem et al. 2022), IBSWO (Mohammad Shtayat et al. 2025) and SAEHO (Sagu et al. 2023), were employed in six studies, proposing outstanding solutions to population-based problems but raising the complexity further due to their usage of advanced operators like IBSWO's flat crossover. Table 12 demonstrates the most effective methodologies.

## 5.2 Emerging algorithms

Modern refinements underscore a growing priority on DL integration, with hybridizations like HAEMPSO-DNN (Saheed et al. 2023) and AAFSO\_GA-FR-CNN (Anushiya and Lavanya 2023) scoring outstanding detection rates surpassing 99%. These methods demonstrate DL's capability to address complex IIoT data patterns affirmatively. Meanwhile, real-time performance aspects remain largely unaddressed, with only a few studies, such as Fang et al. (2024), the binary Genetic Algorithm for Industrial Control Systems, embarking on critical latency necessities. This gap underscores the necessity for more profound research into lightweight, time-efficient methods produced for IIoT environments where rapid attack

**Table 12** Most effective MFS methodologies for intrusion detection

| Methodology                            | Accuracy range | Strength                      | Weaknesses                            |
|----------------------------------------|----------------|-------------------------------|---------------------------------------|
| PSO + RF                               | 99.5–99.7%     | Robust, Generalizable         | Computationally heavy                 |
| Metaheuristic Optimization + XGBoost   | 99.2–99.7%     | High precision, low FPR       | Requires hyperparameter tuning        |
| Hybrid SI (E.g. IBSWO, GTO-BSA, SAEHO) | 95–99.9%       | Cross-dataset reliability     | Complex implementation                |
| DL with Metaheuristics                 | 97–100%        | Handles high-dimensional data | Black-box nature, high resource needs |

detection is critical. The field appears balanced to develop more refined hybrid models that merge the pros of metaheuristics and DL while addressing practical application restrictions.

### 5.3 Critical GAPS

- **Dataset diversity:** 13.9% of these methodologies use general IoT datasets and 50.0% use IoT datasets with industrial relevance rather than IIoT-specific datasets, which decreases the relevance of the fieldwork to IIoT.
- **Computational Efficiency:** Only 19.4% of the case studies discuss the complexity of time in detail or acknowledge high complexity, while 13.9% claim to be lightweight and 66.7% do not detail it enough, despite the real-time needs of IIoT.
- **Real-World Validation:** The majority of papers lack application in live IIoT environments.

### 5.4 Enhanced feature selection

This subsection summarized the outstanding feature selection algorithm, highlighting algorithmic advances and key tradeoffs.

#### 5.4.1 Algorithmic advancements

Contemporary studies underscore hybrid metaheuristics to overpower the limitations of standalone FS methods. These are mostly either

- **SI Hybrids Algorithms** that combine two or more swarm intelligence algorithms to leverage their power. A good example is LGTO-BSA Kareem et al. (2022), which combines Gorilla Troops Optimizer (GTO) with Bird Swarm Algorithm (BSA) to improve exploration, achieving accuracy of 95–98% across data sets. Another example is IB-SWO (Mohammad Shtayat et al. 2025); this hybrid integrates Spider Wasp Optimizer (SWO) with Genetic Algorithm (GA) operators, reducing false positives by  $1.2 \times$  compared to vanilla SWO.
- **Evolutionary-DL Fusion** These are hybridized evolutionary algorithms (EAs) with DL techniques to enrich model performance. One notable instance is HAEMPSO (Saheed et al. 2023), which pairs PSO with autoencoders for nonlinear feature extraction, improving DNN detection rates to 99.9% on UNSW-NB15.
- **Novel Hybrids** the case study (Maazalahi and Hosseini 2025) Uniquely hybridized a metaheuristic, PSO, with clustering's K-means algorithm for intrusion detection.

#### 5.4.2 Key trade-offs

Different metaheuristic algorithm classes exhibit distinct trade-offs in feature selection for IIoT intrusion detection. Table 13 summarizes these trade-offs with examples from recent literature.

Hybrid algorithms like GTO-BSA and ITSO-HGS generally achieve superior accuracy but at the cost of computational complexity. Swarm intelligence approaches (PSO, WOA)



**Table 13** Trade-offs of different metaheuristic fs algorithm typeS

| Algorithm class         | Strengths                                                                            | Limitations                                                                  | Examples                          |
|-------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------|
| Swarm Intelligence      | Good exploration, simpler implementation, high accuracy (e.g., PSO-RF: 99.7%)        | Prone to premature convergence, can be trapped in local optima               | PSO, FFA, AFSA, WOA               |
| Evolutionary Algorithms | Robust global search, handles large feature spaces well, stable convergence          | Computationally expensive, complex parameter tuning                          | GA, IGA, BGA                      |
| Physics-Based           | Well-balanced exploration/exploitation, mathematically well-grounded                 | Slower convergence, highly sensitive to parameter settings                   | BGSA, CVS, FWA                    |
| Human Behavior-Based    | Better at avoiding local optima through memory mechanisms, good for dynamic problems | Complex implementation, often requires domain knowledge                      | TS                                |
| Hybrid SI Algorithms    | Highest accuracy (e.g., GTO-BSA: 95–98%), compensates for individual weaknesses      | Most computationally complex, difficult theoretical analysis, complex tuning | GTO-BSA, AFSA-GOA, SFO-WOA, SAEHO |
| Novel Hybrid approaches | Compensates for individual weaknesses, proposes novel solutions for complex attacks  | Most computationally complex, difficult theoretical analysis, complex tuning | GSGW, IBSWO, GA-PSO               |

**Table 14** Performance Benchmarks

| Model                                      | Accuracy | F1-Score | FPR   | Dataset     |
|--------------------------------------------|----------|----------|-------|-------------|
| GTO-BSA + RF (Kareem et al. 2022)          | 98.2%    | 97.8%    | 0.02% | CICIDS-2017 |
| HAEMPSO-DNN [?]                            | 99.9%    | 99.7%    | 0.01% | BoT-IoT     |
| IBSWO + SVM (Mohammad Shtayat et al. 2025) | 99.7%    | 99.5%    | 0.03% | TON_IoT     |
| Coati-GWO + CVAE (Prasad et al. 2025)      | 99.91%   | 98.11%   | 0.01% | RT-IoT2022  |
| ITSO-BiLSTM-SA (Alblehai 2025)             | 99.44%   | 99.39%   | 0.03% | TON-IoT     |
| IGA-ACBLST (Fang et al. 2025)              | 99.8%    | 99.7%    | 0.02% | TON-IoT     |

offer balanced performance with simpler implementation, while evolutionary algorithms provide robust search capabilities but require careful parameter tuning. The choice of algorithm class depends on specific IIoT deployment constraints, with resource-constrained edge environments favoring lightweight swarm techniques and high-security applications benefiting from hybrid approaches despite their complexity.

Table 14 indicates the most notable performance benchmarks from this review.

## 6 Overall findings

This section outlines the conclusions deduced from the previous sections, analyzing the application of metaheuristic algorithms in IDS for the IIoT, their usefulness in industrial scenarios, and recognizing trends, inconsistencies, and the need for further research, and therefore, it provides a broad overview of the contemporary state of metaheuristic algorithms in IDS.



## 6.1 Analysis metaheuristic algorithms

### 6.1.1 Dominant trends

The case studies generally reveal that the combination of MFS algorithms and classifiers for ID in IIoT networks is a dominant trend, as 34/36 case studies included a form of FS-ML or FS-DL. In 30/36 cases, the MFS was executed first, and then it was followed sequentially by the ML or DL classifier. In the remaining 4, the MFS and ID were hybridized into a single algorithm, while 2 cases did not specify which ID was utilized.

10/33 case studies combined two or more MFS algorithms (e.g., (Dey et al. 2023), (Kareem et al. 2022), (Mohammad Shtayat et al. 2025), and (Gong et al. 2025)). As for ML and deep learning, ML was only a hybrid in ID twice, once with a DL CNN model and once with a metaheuristic PSO (Maazalahi and Hosseini 2025; Sharma and Singh 2025). Meanwhile, DL was hybridized with other DLs in 4/36 case studies and was hybridized with metaheuristics three times (Anushiya and Lavanya 2023) (Taher et al. 2023) (Gong et al. 2025) and once with the aforementioned ML case.

18/36 papers use purely DL for ID, while 3 cases use DL with other ID types. 12/36 case studies use individual ML algorithms, with only two hybrid with DL and metaheuristics. All in all, the trends in terms of classification after MFS in IIoT seem to be dominated by deep learning classifiers and hybrids.

### 6.1.2 Limitations of metaheuristics

As we reviewed the papers, we noticed three main limitations:

- Metaheuristics are black-box, as solely a few papers clarify why certain features were selected, which decreases trust in critical IIoT systems.
- Many studies suffer from overfitting because they convey high accuracy (e.g., 99.9% on BoT-IIoT) but use small and curated datasets that may not generalize well to the real world and therefore more robust evaluation strategies are needed.
- Runtime performance is only evaluated in 9/33 case studies, but real-time processing such as <1 ms latency for PLCs is mandatory for IIoT environments, and therefore better research is needed to improve the complexity of metaheuristics.

## 6.2 Analysis industrial application

### 6.2.1 Practical feasibility

- Dataset Mismatch: 63.9% of the studies (23/36) use general IoT datasets such as NSL-KDD (Zhao 2022) and IoTID20 (Ullah and Mahmoud 2020) or IoT with some industrial relevance such as ToN\_IoT (Alsaedi et al. 2020), BoT\_Iot (Koroniotis et al. 2018), and CICIDS-2017 (Panigrahi 2025). While only 36.1% (13/36) evaluate IIoT-specific data such as WUSTL-IIOT-2021 (Zolanvari 2021), RT-IIoT-2022 (Sharmila and Nagapadma 2023), and SWaT/WADI testbeds (iTrust SUoT 2017, 2016).
- Hardware Constraints: None of the reviewed studies test on edge devices (e.g., Rasp-

berry Pi), though several methods including CVS-FLN (Geetha et al. 2024), PSO-RF (Gaber et al. 2023), and hybrid FFA (Ogunseyi and Thiyagarajan 2025) claim lightweight designs without hardware validation.

- **Attack Coverage Gap:** Most papers focus on conventional network attacks like DDoS (16–21/36) and scanning attacks (10–15/36), while only 3–4 papers address IIoT-specific threats (Gaber et al. 2023) (Fang et al. 2024) (Mohammad Shtayat et al. 2025). Industrial protocol exploits such as Modbus/TCP and DNP3 are largely ignored despite their critical importance in industrial environments.

### 6.2.2 Barriers to adoption

**Proprietary Protocols:** Industrial systems such as PROFINET necessitate protocol-aware IDS, but only 2 papers (Gaber et al. 2023) work around this.

**Regulatory Compliance:** No study aligns with IEC 62443 or NIST SP 800-82 standards for IIoT security.

### 6.2.3 Pathways to real-world deployment

**Co-Design with Industry:** Partner with manufacturers to collect real IIoT traffic from smart factories.

**Edge-Native Algorithms:** Optimize metaheuristics for MCUs such as ARM Cortex-M using quantization.

**Benchmarking Frameworks:** Develop IIoT-designated evaluation metrics such as latency-per-feature.

## 6.3 Open challenges

The final deduced open challenges are as follows:

- **Real-time applicability:** Merely 12/36 papers (Fang et al. 2024) address latency for IIoT applications.
- **Dynamic Data Adaptation:** Most methods presume static feature spaces, disregarding IIoT's streaming data.
- **The usage of IIoT-specific datasets** rather than general IoT-purposed

## 7 PC specifications and software

The PC specifications used to conduct this review are as follows:

- Processor 12th Gen Intel(R) Core(TM) i7-12650 H (2.30 GHz)
- Installed RAM 16.0 GB
- System type: 64-bit operating system, x64-based processor

Python was utilized to generate the figures (Van Rossum and Drake 2009), specifically matplotlib (Hunter 2007). The taxonomy, however, was built using (Miro 2023). The manuscript was written in Lamport (1986) using Digital Science UK Limited (2023).

## 8 Conclusion

In this systematic literature review, we case-studied 36 recent IDS approaches based on MFS usages for ID in IIoT. We discovered that SI algorithms dominate the field, although the most utilized algorithm overall is the evolutionary GA with

5 usages, against 4 usages at most in SI for PSO. Other trending SI algorithms in this field included FFA and GWO.

DL dominates the field, with CNN, ANN, and RNN being employed 6 times each overall, 10 times independently, and 8 times in hybrids. Random Forest (4/36) remains the dominant ML classifier, with PSO-RF combinations showing consistent accuracy between 95 and 99.5% across datasets. All metaheuristic FS approaches are wrapper-based by nature, with 10/36 using hybrid MFS algorithms. Notable high performers include IBSWO, Coati-GWO-CVAE, FFA-SPAARC and HAEMPSO-DNN, all achieving >99.7% accuracy. We identified key trade-offs across metaheuristic classes: swarm intelligence algorithms offer simpler implementation but risk premature convergence, while hybrid approaches achieve superior accuracy at the cost of computational complexity. Critical gaps included: (1) mismatch of the data set (13.9% using generic data and 50% using IoT with industrial relevance datasets rather than IIoT-specific data), (2) limited coverage of the attack (only four articles addressing industrial-specific threats), (3) lack of edge device validation, and (iv) insufficient evaluation of real-time performance (only four articles). Those gaps also present feasibility problems in implementing the models of the case studies, which leaves some open challenges to the readers, including real-time applicability of these models and the usage of IIoT-specific datasets for the specific purposes the authors are building their models. Our future work will address these challenges, particularly focusing on IIoT-specific attack detection and the real-time applicability of metaheuristic feature selection. Our future work can also include creating similar systematic literature reviews, this time for MFS for different usages other than IIoT, like cloud computing and the Internet of Medical Things (IoMT), and also potentially non-metaheuristic FS methods for IIoT, like filtering. Furthermore, it can include a focus on this topic with AI explainability, interpretability, and feature importance. And discuss online/streaming adaptation for FS in IIoT and concepts such as drift handling and incremental/online feature selection.

## 9 Supplementary information

Additional supporting information may be found in the online version of the article at the publisher's website.

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## Declarations

**Conflict of interest** The authors declare they have no Conflict of interest.

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**Code availability** Not applicable.

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